

6 Obesity increases the risk of cardiovascular disease (CVD).

Body mass index (BMI), waist-to-hip ratio (WHR) and skinfold thickness are indicators of obesity.

- (a) The table shows some measurements taken from two females, female **J** and female **K**.

Female	Height / cm	Mass / kg	Waist / cm	Hips / cm	BMI	WHR
J	155	59.1	80	100	25	0.80
K	155			125	36	0.80

- (i) Calculate the waist size of female **K**.

(1)

Answer cm

- (ii) The formula for calculating BMI is:

$$\text{BMI} = \frac{\text{mass in kg}}{(\text{height in m})^2}$$

Calculate the mass of female **K**, using the data in the table.

(3)

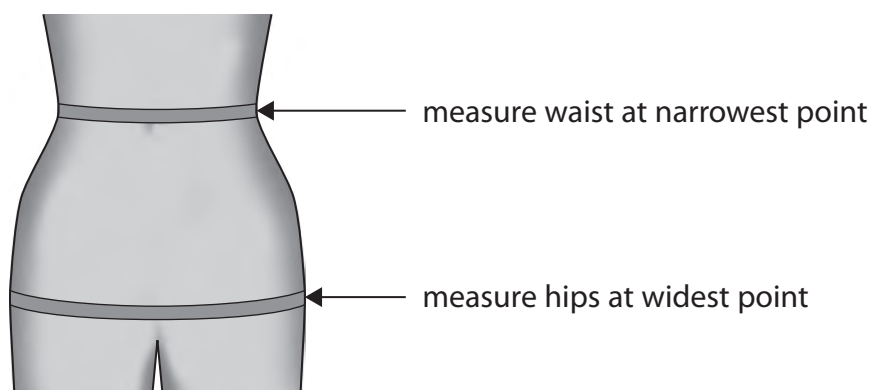
Answer kg



(iii) Comment on the risk of developing CVD in these two women.

(2)

(iv) The diagram shows how the waist and hip measurements should be taken, using a tape measure.

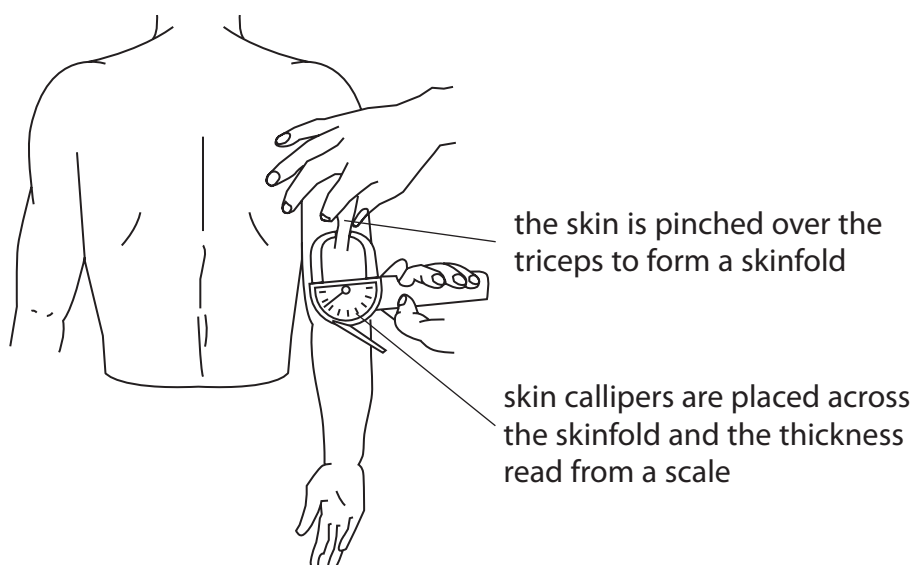


Explain how the way a person takes these measurements could produce an underestimate of their risk of CVD.

(2)



- (b) The diagram shows how a skinfold thickness measurement is taken over the triceps muscle at the back of the arm.



Skinfold thickness measures the thickness of the layer of fat under the skin.

Measurements are taken from several sites on the body.

The table shows the skinfold thickness at four sites on a 42-year-old female.

Site	Skinfold thickness / a.u.
over the biceps at the front of the arm	0.63
over the triceps at the back of the arm	0.82
under the shoulder blade at the back of the neck	0.65
above the hip bone at the side of the body	0.82

- (i) Suggest **two** reasons why the skinfold thickness values are different at each site on the body.

Assume that the skin callipers have been used correctly.

(2)



- (ii) The table shows a body fat interpretation chart.

The values in the table are the means of the four skinfold thickness measurements.

Age	Level of body fat			
	Low	Moderate	High	Very high
20 to 29	<0.71	0.71 to 0.77	0.78 to 0.82	>0.82
30 to 39	<0.72	0.72 to 0.78	0.79 to 0.84	>0.84
40 to 49	<0.73	0.73 to 0.79	0.80 to 0.87	>0.87
50 to 59	<0.74	0.74 to 0.81	0.82 to 0.88	>0.88

Determine the level of body fat of this female.

(1)

Answer

(Total for Question 6 = 11 marks)

